计算系统生物学作业3

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2026 年 4 月 8 日

Feed-forward loop network motif

1. The four-node diamond pattern occurs when X regulates Y and Z, and both Y and Z regulate gene W.

(a) In a random ER network, we set the mean connectivity to k. This means that the probability that there exists a directed edge is $p = \frac{k}{N}$.

To form a diamond pattern, two conditions have to be satisfied.

(1) choose 4 nodes. Notice that Y and Z are symmetric, we have $N \cdot (N-1) \cdot \frac{(N-2)(N-3)}{2}$ kinds of choices.

(2) there exists 4 certain edges(X→Y,X→Z,Y→W,Z→W). The probability is $p^4 = (\frac{k}{N})^4$

$\mathbb{E}[N_d] \propto N^4 \cdot (\frac{k}{N})^4 = k^4$

This implies that the mean number of diamonds is not related to N, but is merely determined by mean connectivity k.

(b) With 2 possibilities for each edge(+ or -), we have $2^4 = 16$ kinds of combinations. If we consider the symmetry of Y and Z, there are 10 kinds of combinations.

A pattern is called coherent, if and only if the route from X to W has the same overall sign.

We consider the product of the route sign. X→Y→Z

(+)=(+,+) or (-,-), $2 \times 2 = 4$

(-)=(+,-)or (-,+),$2 \times 2 = 4$

Hence we have 8 that are coherent(6 if we consider symmetry).

(c) Assume the threshold level is $K_{XY}, K_{XZ}, K_{YW}, K_{ZW}$.

Assume that Y responds more quickly than Z.

ON delay: $t_{Y,on} < t_{Z,on}$

OFF delay: $t_{Y,off} < t_{Z,off}$

AND logic:

ON: delay=max$(t_{Y,on}, t_{Z,on}) = t_{Z,on}$

OFF: delay=min$(t_{Y,off}, t_{Z,off}) = t_{Y,off}$

Short on signals might be ignored.

OR logic:

ON: delay=min$(t_{Y,on}, t_{Z,on}) = t_{Y,on}$

OFF: delay=max$(t_{Y,off}, t_{Z,off}) = t_{Z,off}$

W is sustained despite an interrupted input signal.

There is no sign-sensitive delay, unless the responding rate of Y and Z are very different.

2.Solve tge dynamics of the Type-3 coherent FFL with AND logic at the Z promoter in response to steps of S_x.

X activates Y, X inhibits Z, Y inhibits Z. Z=(NOT X*) AND (NOT Y*)

When S_X is turned on, X quickly turns into X*, which binds the Z promoter immediately, and the production of Z will stop at once. There is no apparent delay.

When S_X is turned off, X* disappeared quickly, but the concentration of Y is still high and will keep binding the Z promoter. Hence OFF to ON of Z is delayed, its time determined by the degrading rate of Y.

If $S_X = 0$, X* is low, Y* is low, Z=ON.

If $S_X = 1$, X* is high, Y* is high, Z=OFF.

Type-1 cFFL: AND($S_X \wedge S_Y$),OR(S_X)

Type-2 cFFL: AND($\bar{S}_X \wedge S_Y$),OR($\bar{S}_X$)

Type-3 cFFL: AND($\bar{S}_x$),OR($\bar{S}_X \wedge \bar{S}_Y$)

Type-4 cFFL: AND(S_X),OR($S_X \vee \bar{S}_Y$)

3.Consider a coherent type-1 FFL with nodes X,Y and Z, which is linked to another coherent type-1 FFL in which Y activates Y_1, which activates Z

(a) We have two nested Type-1 cFFL, (X,Y,Z) and (Y,Y_1,Z), and all of them play the function of activation.

Here we assume that the pattern satisfies AND logic.

There is no delay in all OFF type since X disappears immediately under AND logic, which binds Z promoter. Hence we only consider ON below.

When there are S_X, S_Y and no S_{Y1}, there is a lot of X*,Y*,Y1. When S_{Y1} becomes 1, there is no delay, hence no sign-sensitive delay.

When there are S_X, S_{Y1} and no S_Y, there is a lot of X*,Y. When $S_Y = 1$, Y is activated to Y*, and then promotes the production of Y1*, and when Y1* reaches threshold, Z starts to be produced. Hence there is delay, and sign-sensitive delay.

When there are S_Y, S_{Y1} and no S_X, we only have a lot of X. When S_X becomes 1, similarly there is a delay. Since there are more steps, there is a longer delay and the sign-sensitive delay.

The two systems cannot be separated, since in the last situation above, we need to compare the threshold and production rate to determine the T_{delay}

(b) To make it easier for our discussion, we quantify the promotion/inhibition by X to Y:1, Y1 to Z: 0.5, Y to Z:-1.

(i)When there are S_X, S_Y and no S_{Y1}, there is a lot of X*,Y*,Y1, the production rate of Z is 0. When S_{Y1} becomes 1, Y1 alleviates the regression of Y to Z, setting the production rate to 0.5, and there is no delay.

(ii)When there are S_X, S_Y, S_{Y1}, there is a lot of X*,Y*,Y1*, the production rate of Z is 0.5.

1.When S_{Y1} becomes 0, Y1 is deactivated at once, and the production of Z is inhibited, and there is no delay.

2.When S_Y becomes 0, Y* is deactivated at once, and the production rate of Z is 1.5, which decreases to 1 gradually. There is no delay.

3.Wehn S_X becomes 0, X* is deactivated, and the production rate of Z is -0.5. Eventually there is no promoting material. Hence the production rate of Z returns to 0. There is no delay.

(iii) When there is S_X, S_{Y1} and no S_Y. There is a lot of X*,Y and no Y1. When S_Y becomes 1, Y is immediately activated to Y*, which simultaneously inhibit the production of Z and promote the production of Y1*, and the production rate of Z is 0. After a while, Y1* starts to alleviate the regression, and the production rate of Z is down to 0.5.

(iv) When there are S_Y, S_{Y1}, and no S_X, there is only a lot of X, and no Y,Y1. When S_X turns to 1, X is activated to X* at once, and the production rate of Z is 1. The situation is divided into 2 categories.

1.Y^* reaches its threshold first. The production rate of Z is first inhibited (to 0), and then alleviated to 0.5, with no delay.

2.$Y1^*$ reaches its threshold first. The production rate of Z is first promoted (to 1.5) and then inhibited to 0.5, with no delay.